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# Multimodality as Bridge or Hurdle in Human–AI Communication

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- Humans often infer more than the literal content of an image; what does it communicate?
- Meaning may depend on context, shared knowledge, and abstraction; AI often only captures surface associations

**Example:**

- Literal description: *“two children in a park”*
- Underlying meaning: *friendship*

**We want to observe what role multimodality plays in communication.**

Figure: Literal scene, but also a possible abstract meaning.

# Core Question



- When does multimodality help communication?
- When does it make communication harder?
- What expectations do people have regarding AI capabilities in the context of multimodal communication?

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**Multimodality as bridge**

- one modality compensates for another
- helps when a concept is hard to express in words alone
- can support abstraction and communication

**Multimodality as hurdle**

- additional meaning has to be inferred
- context and association matter
- sender and receiver may align differently

# Why Games?



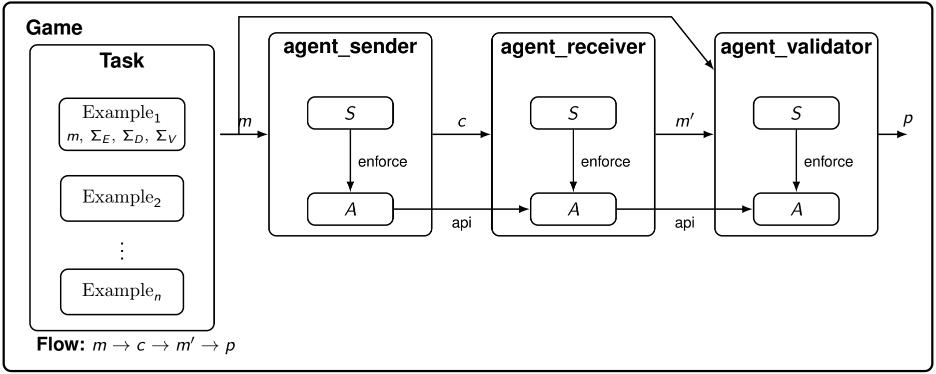
- Many **multimodal communication tasks** are gamelike and follow a similar framework
- They separate three roles:
  - **sender**: encodes a concept
  - **receiver**: decodes it
  - **verification**: checks success


**Why useful here?**

- this abstraction captures a large variety of games
- the "difficulty", serves as bridge/hurdles

# Framework Overview






**The framework separates the task definition, agent behavior, interface constraints, and round-level control flow.**

# Alphabets and Signatures introduce difficulty

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**Alphabets**
$\Sigma_E$: encoder alphabet, $\Sigma_D$: decoder alphabet
En/Decode with emojis/ images rather than letters

**Signatures define valid outputs**
single choice: $x \in \Sigma$, multiple choice: $X \subseteq \Sigma$, word sequence: $\Sigma^+$
More restriction on choice, limits en/decoder

**We make games hard, to see change in en/decoder behavior under different conditions**

# Expectations regarding AI capabilities

—

# Via captioning task

- We instantiate the framework with a captioning game over **image pairs**
- For each pair, a human encoder selects one image as the target message $m$
- The encoder then produces a caption $c$ that should help a receiver identify the target

**Two conditions**

1. **Standard captioning:** write a caption so that another human can identify the target image
2. **AI-aware captioning:** write a caption that remains understandable for humans, but should be harder for AI to decode

**Assumption:** Images have many hidden meaning; human might have expectations what other humans might decode but AI doesn't

- **30 images** organized into **15 visually related pairs**
- Three categories:
  - **Internet memes** (cultural knowledge, humor, implicit context)
  - **Stock photography** (literal and prototypical concepts)
  - **Abstract art** (weak object-level semantics, more interpretive abstraction)

**Pairing strategy**

- **Abstract art:** pairs from the **same artist**
- **Stock images:** photos from the **same search term**
- **Memes:** paired by **close CLIP embeddings**

**Goal:** create pairs with different types of similarity and ambiguity to test selective interpretability.

# Live Demo!

# Evaluation Setup



| | Human raters | AI raters | Human - AI |
|---|---|---|---|
| **baseline** | Fleiss' $\kappa$ | Fleiss' $\kappa$ | Cohen's $\kappa$ |
| **AI-aware** | Fleiss' $\kappa$ | Fleiss' $\kappa$ | Cohen's $\kappa$ |
| **accuracy** | Accuracy vs. ground truth | Accuracy vs. ground truth | |

## Expectation

- AI-aware condition should make decoding harder for both humans and AI
- Humans should remain more accurate and consistent than AI
- The gap may reflect how well humans anticipate AI's capabilities

# Main Results



| Metric | Standard | AI-aware |
|---|---|---|
| Human Fleiss' $\kappa$ | 0.954 | 0.636 |
| AI Fleiss' $\kappa$ | 0.591 | 0.679 |
| Human–AI Cohen's $\kappa$ | 0.167 | -0.024 |
| Mean human accuracy | 98.3% | 85.0% |
| Mean AI accuracy | 52.2% | 43.3% |
| AI majority accuracy | 53.3% | 40.0% |

- Harder captions hurt **both** humans and AI
- But they also increase **human–AI divergence**

**The hurdle changes not only performance, but interpretation.**

# What Changed in the AI-aware Condition?



**Humans**

$\kappa$

$0.954 \rightarrow 0.636$

- shorter captions
- less literal
- more ambiguous

**AI**

accuracy

$52.2\% \rightarrow 43.3\%$

- fewer object cues
- more abstract wording
- harder to anchor

**The captions become less literal — and harder to decode.**

# Humans and AI Drift Apart



**AI self-agreement**

$\kappa$

$$0.591 \rightarrow 0.679$$

- not random failure
- more similar mistakes

**Human–AI agreement**

$\kappa$

$$0.167 \rightarrow -0.024$$

- different cues
- different interpretations

**AI becomes less human-aligned, not simply more random.**

# How Do Humans Encode Against AI?

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**Standard**

*[Forest and grass with sunshine, some shadow in the foreground and in the background more dense forest]*

longer, literal, descriptive

**AI-aware**

*[Fairy Tale]*

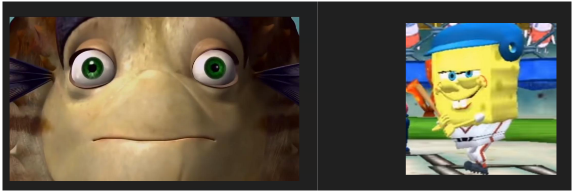
shorter, indirect, evocative

**Humans seem to shift from scene description to feeling, metaphor, or association.**

# AI Priors Can Override the Caption



- familiar concepts can dominate
- AI may choose what it already "knows"
- caption matters less than prior salience



**Example:**
*SpongeBob*

**Reading:**
stable, but biased decoding

# Limitations



- Small dataset and participant number
- Ground truth serves as both reference label and part of the human evaluation setup
- Ai ratings are repeated runs from the same predated model
- Agreement metrics such as $\kappa$ are sensitive to label prevalence

# Conclusion



- Framwork that captures multimodal games
- AI-aware captions made the task harder for everyone
- Humans still outperformed AI
- Human–AI agreement became even weaker
- Humans used more indirect captions:
  - less object-based
  - more mood, metaphor, association

**The hurdle reduced success,**
**and also pushed humans and AI toward different interpretations.**